



Immunogenicity and safety of Biological E's 14-valent pneumococcal conjugate vaccine (PNEUBEVAX 14®) administered in a 2p + 1 schedule to healthy infants: a multicenter, randomized, active controlled, single-blind, phase III trial

Subhash Thuluva,^{a,*} Ramesh V. Matur,^a Subbareddy Gunneri,^a Siddalingaiah Ningaiah,^a Vijay Yerroju,^a Rammohan Reddy Mogulla,^a Chirag Dhar,^a Kamal Thammireddy,^a Piyush Paliwal,^a Anand Kawade,^b Pradeep Nanjappa,^c Jog Pramod,^d Manish Narang,^e Bheemisetty S. Chakravarthy,^f and Prashanth M. Virupakashappa^g



^aBiological E Limited, Hyderabad, India

^bKing Edward Memorial Hospital Research Centre, Pune, Maharashtra, India

^cCheluvamba Hospital, Mysuru, Karnataka, India

^dMedipoint Hospital, Pune, Maharashtra, India

^eUniversity College of Medical Sciences & Guru Teg Bahadur Hospital, Delhi, India

^fKing George Hospital, Visakhapatnam, Andhra Pradesh, India

^gMedstar Specialty Hospital, Bengaluru, Karnataka, India

Summary

Background Pneumococcal conjugate vaccines (PCVs) have markedly reduced childhood pneumococcal diseases, yet serotype replacement and regional heterogeneity remain important challenges. The World Health Organization recommends either a 3p + 0 or 2p + 1 schedule for PCV immunization programmes. BE-PCV14, a 14-valent vaccine, has previously been shown to be non-inferior to PCV13 in a 3p + 0 regimen. In this Phase III trial, we aimed to descriptively compare the immunogenicity and safety of BE-PCV14 and PCV13 in a 2p + 1 schedule in Indian infants.

Methods In this randomized, single-blind, multicenter trial, 400 PCV-naïve infants (6–8 weeks old) were randomized 1:1 to receive either BE-PCV14 or PCV13; at 6 and 14 weeks, with a booster at 9 months. Serum IgG against 14 vaccine serotypes plus cross-protective 6 A were measured at post primary (28 days post dose 2), pre booster (at 9 months) and post booster (30 days post dose 3) time points. The primary endpoint was the proportion achieving IgG ≥ 0.35 $\mu\text{g/mL}$ (seroresponse rate) for the 12 serotypes common to both vaccines at post primary, pre booster and post booster time points. Solicited local and systemic reactions were recorded for 7 days after each dose; unsolicited, and serious adverse events (SAEs) were captured throughout.

Findings Between May 2023 and July 2024, 400 participants were enrolled of which 380 (95%) completed the study. Post primary seroresponse rates in the BE-PCV14 arm for common serotypes ranged from 72.6% (95% CI: 65.8, 78.5) (serotype 3) to 100% (95% CI: 98.0, 100.0) (14, 19 F); PCV13 rates ranged from 71.6% (95% CI: 64.9, 77.5) to 100% (95% CI: 98.1, 100.0) (14, 19 F, 19 A). Post booster rates were 87.6% (95% CI: 82.1, 91.6) to 100% (95% CI: 98.0, 100.0) for BE-PCV14 and 85.0% (95% CI: 79.4, 89.4) to 100% (95% CI: 98.1, 100.0) for PCV13. BE-PCV14 elicited high responses against the two additional serotypes, i.e., 22 F: 96.8% (95% CI: 93.1, 98.5), 33 F: 92.5% (95% CI: 87.8, 95.5), and cross-protective 6 A: 93.0% (95% CI: 88.4, 95.9). Of the participants that received BE-PCV14 or PCV13, 33.5% (95% CI: 27.3, 40.3) and 38% (95% CI: 31.6, 44.9) had mild AEs and 11.5% (95% CI: 7.8, 16.7) and 10.5% (95% CI: 7.0, 15.5) had moderate AEs, respectively. Two unrelated SAEs occurred in the BE-PCV14 arm.

Interpretation Administered in a 2p + 1 schedule, BE-PCV14 was highly immunogenic, well tolerated, and comparable to PCV13 while broadening serotype coverage. These findings support consideration of BE-PCV14 for routine infant immunization programmes using a 2p + 1 schedule, particularly in settings where the additional serotypes contribute to ongoing pneumococcal disease.

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*Corresponding author. Clinical Development, Biological E. Limited, 18/1&3, Azamabad, Hyderabad, 500 020, Telangana, India.

E-mail address: Subhash.Thuluva@biologicale.com (S. Thuluva).

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Keywords: Pneumococcal conjugate vaccine; 2p + 1; Immunogenicity and safety; BE-PCV14; Infants

Research in context

Evidence before this study

We searched PubMed/MEDLINE, Embase, and Cochrane CENTRAL, and reviewed reference lists of relevant publications, from database inception to December, 2025, without language restrictions, using terms for pneumococcal conjugate vaccines and schedules (including “pneumococcal conjugate vaccin*”, “PCV”, “PCV13”, “PHiD-CV”, “Synflorix”, “Pneumosil”, “SIIP-PCV”, “BE-PCV14”, “Biological E”, “2 + 1/“2p+1”, “booster”, “immunogenicity”, and “safety”). We also searched medRxiv for relevant preprints over the same period. We included clinical studies in infants/young children reporting immunogenicity and/or safety outcomes and excluded non-clinical studies and studies limited to adults. We did not undertake meta-analysis because schedules, endpoints, and reporting were not sufficiently comparable; we considered key methodological limitations qualitatively. The evidence supports the immunogenicity, effectiveness, and safety of PCV13 and PHiD-CV administered as 2 + 1 schedules, and SIIP-PCV (Pneumosil) was licensed and WHO-prequalified following demonstration of immunological non-inferiority and acceptable safety in infant studies. BE-PCV14, which adds serotypes 22 F and 33 F, has shown robust immunogenicity and acceptable safety and has previously met non-inferiority criteria vs PCV13 in 3p + 0 studies; a Phase IV 3p + 1 evaluation has been reported as a preprint. However, we identified no peer-reviewed clinical data evaluating BE-PCV14 in a 2p + 1 schedule in Indian infants, providing the rationale for this Phase III study.

Added value of this study

To our knowledge, this randomized Phase III trial is the first to evaluate BE-PCV14 administered in a WHO-recommended 2p+1 schedule in Indian infants and to directly compare it with PCV13. Immune responses to the 12 shared serotypes were generally comparable between groups after the primary series and after the booster, with a strong booster response following BE-PCV14. BE-PCV14 also induced robust responses to the additional serotypes 22 F and 33 F, and responses to serotype 6 A were observed, consistent with cross-reactivity associated with inclusion of serotype 6 B. In post hoc analyses applying non-inferiority criteria, BE-PCV14 met non-inferiority vs PCV13 based on seroresponse rates and IgG geometric mean concentrations. Reactogenicity and overall safety were similar between vaccines, with no new safety signals identified.

Implications of all the available evidence

Administered as a 2p+1 schedule, BE-PCV14 maintains coverage of PCV13 serotypes while expanding it to serotypes of current epidemiological relevance, 22 F and 33 F. Its comparable immunogenicity and safety profile to PCV13 supports its programmatic use in routine infant immunization schedules. Together with global evidence supporting the immunogenicity and effectiveness of 2p+1 PCV schedules, these findings support the consideration of BE-PCV14 within national infant programs, including in countries across the Global South that have adopted 2p+1 regimens.

Introduction

Streptococcus pneumoniae remains a major cause of pneumonia, meningitis and bacteraemia in children under 5 years of age, with low- and middle-income countries (LMICs) accounting for most of the estimated 672,000 annual deaths.^{1,2} The introduction of 7-, 10- and 13-valent pneumococcal conjugate vaccines (PCVs) into infant programmes has drastically reduced vaccine-type invasive pneumococcal disease (IPD) and pneumonia.^{3–5} However, this success has been accompanied by serotype replacement, with non-vaccine serotypes such as 22 F and 33 F emerging as significant causes of residual infections in post-PCV13 settings across multiple regions.^{6,7} Indian data likewise list 22 F and 33 F among the ten most common non-vaccine serotypes causing paediatric IPD.^{8,9} The inclusion of these serotypes in newer PCVs such as Biological E’s

PNEUBEVAX 14®(BE-PCV14), PCV15 and PCV20, is aimed at addressing this evolving serotype distribution and preventing IPD related to these additional serotypes.^{10,11}

The World Health Organization recommends either a three-dose primary series (3p+0) or two-dose primary series with a booster dose (2p+1) for PCVs. The 2p+1 schedule consists of primary doses at 6 weeks and 14 weeks followed by a booster dose at 9–15 months of age.¹² While both schedules are highly effective, a 2p+1 regimen induces higher antibody concentrations in the second year of life that may enhance herd protection while requiring fewer infant visits.^{13,14} More than 50 countries worldwide have now adopted the 2p+1 schedule in their routine vaccination programmes.¹⁵ BE-PCV14 has undergone extensive clinical development with a phase I trial (CTRI/2017/06/008759), a

phase II trial (CTRI/2017/11/010387), two phase III trials (CTRI/2020/02/023129 and CTRI/2021/10/037067), and a phase IV study (CTRI/2023/09/057894) demonstrating an acceptable safety profile. The two phase III clinical trials demonstrated immunogenic non-inferiority of BE-PCV14 to Pfizer's PCV13 in a 3p+0 schedule and a comparable safety profile.^{16,17} In the current manuscript, we report the findings of a Phase III trial that evaluated the immunogenicity and safety of BE-PCV14 compared to PCV13, when administered to healthy Indian infants in a 2p + 1 dosing regimen. Though the primary immunogenic objective was to descriptively compare responses after BE-PCV14 or PCV13, a formal *post hoc* non-inferiority analysis was also conducted.

Methods

Study design and participants

This single-blind, randomised, active-controlled phase III trial evaluated the immunogenicity, safety and tolerability of BE-PCV14 (PNEUBEVAX 14[®]) administered in a 2p+1 schedule to healthy Indian infants. The study was conducted at seven sites in India between 15 May 2023 and 5 July 2024 (sites details included as [Supplementary Data, Appendix 1](#)). The protocol (included as [Supplemental Date](#)) complied with the Declaration of Helsinki, Good Clinical Practice and national regulations, and was approved by the ethics committee of Guru Teg Bahadur Hospital, Delhi (ECR/510/Inst/DL/2014/RR-20); King George Hospital, Visakhapatnam (ECR/197/Inst/KGH/2013/RR-20); Mysore Medical College and Research Institute, Mysuru (ECR/134/Inst/KA/2013/RR-19); Sant Dnyaneshwar Medical Education & Research Centre, Pune (EC/RENEW/INST/2024/17666); Medipoint Hospitals Pvt. Ltd., Pune (ECR/357/Inst/MH/2013/RR-20); Medstar Speciality Hospital, Bangalore (ECR/1324/Inst/KA/2019); and KEM Hospital Research Centre, Pune (ECR/272/Inst/MH/2013/RR-22). Written informed consent was obtained from participant's parents or legal guardians before any study procedures. There was no patient or public involvement in designing, conducting, or reporting this study. During the conduct of this study, there were no major protocol deviations reported at any of the study sites. Few subjects reported for their visits, out of window period, but these deviations were not found to be significant. There were no deviations reported, that could adversely affect the safety and well-being of the subjects or the scientific validity of the study. The protocol deviations, pertaining to out of window were communicated to ethics committees of the respective study sites. Also, there were no changes made to the Ethics committee approved trial protocol, including inclusion/exclusion criteria, outcomes, or analysis.

Eligible infants were 6–8 weeks of age, pneumococcal-vaccine naïve, healthy, and with a

minimal vaccination status of birth dose of BCG, OPV and Hepatitis B vaccine. Infants with evidence of previous *S. pneumoniae* infection and/or immunosuppression/on immunosuppressive therapy were excluded. All inclusion/exclusion criteria are listed in [Supplementary Data, Appendix 2](#). A total of 400 infants were enrolled.

Randomization and masking

At the screening (visit 1), all participants were randomized 1:1 to receive either BE-PCV14 or PCV13 (Pneumar 13[®], Pfizer). The randomization was conducted by using an interactive web-response system (IWRS) generated with PROC PLAN (SAS 9.4). The study was single-blind, with parents/legal representatives unaware of the assigned vaccine. The laboratory personnel involved in serology testing remained blinded to the treatment, and codes were used to link participants and samples without revealing treatment.

Procedures

The study comprised five scheduled visits. On Day 0 (Visit 1), participants were screened, randomized, had baseline blood drawn, and received the first dose of either BE-PCV14 or PCV13. The second dose was administered on Day 56 (Visit 2). Post-primary immunogenicity samples were collected on Day 84 (Visit 3). At 9 months of age (Visit 4), pre-booster blood was drawn and a booster dose was administered. Final post-booster blood samples were obtained 30 days after the booster (approximately 10 months of age, Visit 5). Safety-follow-up data were recorded on Days 56 and 84, and at the post-booster timepoints. Each participant was followed until 30 days post-booster with a permissible +7-day window for Visits 2, 3, 4 and 5.

Each 0.5 mL dose in the multi-human dose presentation of PNEUBEVAX 14[®] (BE-PCV14) contained 3 µg of serotype 1 polysaccharide; 2.2 µg of each of serotypes 3, 4, 5, 7 F, 9 V, 14, 18 C, 19 A, 19 F, 22 F, 23 F and 33 F; and 4.4 µg of serotype 6 B, conjugated to 20–50 µg CRM₁₉₇ and adsorbed on ≤ 0.75 mg aluminium phosphate, with 4 mg of 2-phenoxyethanol. PCV13 contained 2.2 µg of each of serotypes 1, 3, 4, 5, 6 A, 7 F, 9 V, 14, 18 C, 19 A, 19 F, and 23 F saccharides; 4.4 µg of 6 B saccharide, conjugated to 34 µg CRM₁₉₇ with 125 µg aluminium phosphate. Both vaccines were administered intramuscularly in the anterolateral aspect thigh.

All participants had received birth dose of BCG, Hepatitis B and Oral Polio vaccines prior to enrolment. During the study, concomitant administration of vaccines recommended in the Universal Immunization Programme (UIP) at 6-10-14 weeks and 9 months schedule was permitted.

Approximately 5.0 mL of peripheral venous blood was collected at four time points: Visit 1, 3, 4, and 5. Blood samples were centrifuged at approximately

3500 rpm for 10–15. The separated serum was transferred into pre-labelled cryovials/serum separation containers using sterile micro-pipettes. The cryovials were stored at $<-20^{\circ}\text{C}$ till transfer to the central laboratory and sponsor facilities. Anti-pneumococcal capsular polysaccharide (PnCPS) IgG concentrations estimation against each of the 14 vaccine serotypes and 6 A was performed at Biological E's in-house laboratory using the World Health Organization (WHO) reference enzyme linked immunosorbent (ELISA) assay¹⁸ with a minor modification. Cell wall polysaccharide (CWPS) multi was used instead of CWPS plus 22 F pneumococcal polysaccharide (PnCPS) for preadsorption to neutralize non-specific antibodies in the coating antigens.

Following each vaccination, infants were observed on-site for a minimum of 60 min to monitor for any immediate adverse reactions and to provide medical management if required. During each vaccination visit, the parent(s) or legally acceptable representative (LAR) received a structured diary card to document any solicited local or systemic adverse events (AEs) occurring within the first 7 days post-vaccination. The list of solicited AEs is provided in [Supplementary Data, Appendix 3](#). In addition to solicited AEs, all unsolicited adverse events, and serious adverse events (SAEs) were recorded throughout the study duration. The intensity of each adverse event was graded using Common Terminology Criteria for Adverse Events (CTCAE) v5.0, and the Division of AIDS (DAIDS) Table for Grading the Severity of Adult and Pediatric Adverse Events, version 2.0. For febrile events, severity was classified according to the Brighton Collaboration case definition scale. Causality assessment was performed by the study investigator, who determined the relationship between each reported event and the study vaccine (investigators assessed causality).

Outcomes

The primary outcome reported from this trial was the proportion of infants with serotype-specific anti-PnCPS IgG $\geq 0.35\text{ }\mu\text{g/mL}$ (defined as the seroresponse rate¹⁹) with exact 2-sided 95% CIs at post-primary, pre-booster, and post-booster timepoints for each vaccine serotype. Other immunogenicity endpoints reported are geometric mean concentrations (GMCs) of anti-PnCPS IgG for each serotype with exact 2-sided 95% CIs. A secondary outcome was the safety, reactogenicity and tolerability of BE-PCV14 in comparison to PCV13, measured by incidence rates of local and systemic adverse events (AEs), and serious adverse events (SAEs).

The clinical trial protocol, finalized in 2020, reflected then-prevailing national regulatory agency-aligned expectations to capture both titre- and concentration-based immunogenicity endpoints. During study conduct and close-out in 2025, regulatory and

laboratory practices evolved toward calibrated concentration ($\mu\text{g/mL}$)-based reporting and use of the established $0.35\text{ }\mu\text{g/mL}$ post-vaccination seroresponse threshold. Accordingly, concentration-based outcomes of seroresponse rates and GMCs were reported. This manuscript also presents concentration-based outcomes to align with the current regulatory guidance and our current clinical serology workflow, in line with reporting of immunogenicity data for recently licensed PCVs.

Sample size determination

This phase III study was designed to descriptively compare the immunogenicity and safety of BE-PCV14 administered in a 2p + 1 schedule with that of Pfizer's PCV13. Although a concurrent comparator arm was included, the trial was not formally powered to demonstrate statistical differences between the test and control arms. With 400 participants randomized evenly between the two vaccine groups, the design aimed to generate robust descriptive data to support clinical and programmatic decision-making. This number was based on established precedent from prior PCV trials and current regulatory guidance.^{20,21}

Statistical analyses

All demographic and baseline characteristics for each treatment group were summarized in all randomized participants defined as the intention-to-treat (ITT) population. All safety analyses were descriptive and performed for all those who received at least one dose of the vaccine (safety population). Demographics, and baseline characteristics were analyzed by summary statistics. For continuous variables, n, mean, standard deviation, median, and range (minimum and maximum) were presented. For categorical data, frequencies were computed. All reported adverse events during the entire study period were summarized by calculating frequencies and were listed per subject including severity and relationship to the vaccine (causality).

The number and percentage of participants with AEs were presented by system organ class (SOC) and preferred term (PT). Two-sided 95% exact confidence intervals (CIs) were calculated for occurrence rates of reported AEs and SAEs during the study. All AEs were coded using the Medical Dictionary for Regulatory Activities (MedDRATM; version 25.1) coding dictionary, and concomitant medications were coded using the WHO Drug Dictionary, March 2022.

Immunogenicity analyses were performed in the per-protocol (PP) population (participants who received all doses, had no major protocol deviations and contributed valid immunogenicity data at all protocol defined timepoints) at Day 84 (post-primary), 9 months (pre-booster) and at 10 months (post-booster). The protocol allowed for visits that occurred within a

window of 7 days after the due date to promote compliance. Seroresponse rates and geometric mean concentrations (GMCs) were calculated for each vaccine serotype along with their two-sided 95% CIs. The Wilson score method, without continuity correction, was used for seroresponse rates and natural log transformation was used for GMC calculations, and the results were back-transformed. Reverse cumulative distribution (RCD) curves of anti-PnCPS antibody concentrations by serotype for post-primary and post-booster timepoints were plotted for all common vaccine serotypes in both groups.

Immune responses to serotypes 22 F and 33 F (not present in the PCV13 group) after BE-PCV14, were assessed in terms of seroresponse rates and geometric mean concentrations in comparison with the lowest performing serotype in the PCV13 group as a reference at each timepoint.

Missing data were not imputed, analyses were performed using available (observed) data only. Participants who discontinued were considered discontinued, and no replacement of missing values was undertaken. Immunogenicity analyses were performed in the per-protocol population comprising evaluable participants with immunogenicity data available at all protocol-defined timepoints, participants with missing immunogenicity data were excluded from per-protocol summaries. Safety analyses included all vaccinated participants, and percentages were calculated using the number of observed values as the denominator, where applicable.

Post hoc non-inferiority analyses were performed to compare the final (post-booster) responses of BE-PCV14 with PCV13. The 95% CI for the difference in seroresponse rate was calculated for each vaccine serotype by the Farrington-Manning method. Non-inferiority was to be concluded if the lower limit of the 95% CI was $\geq -10\%$. Similarly, post-booster GMC ratio and 95% CI (log-transformed and back transformed) of BE-PCV14 to PCV13 was also calculated for each vaccine serotype. Non-inferiority was to be concluded if the lower limit of the 95% CI of the ratio was >0.5 .

Role of the funding source

The study was funded by Biological E. Limited. The funder was involved in the study design, collection, analysis, interpretation of data, the writing of this article, and the decision to submit it for publication.

Results

The study was conducted at seven sites in India between 15 May 2023 and 5 July 2024. A total of 400 infants aged 6–8 weeks were screened and randomized to receive either BE-PCV14 ($n = 200$) or PCV13 ($n = 200$). The vaccines were administered as described earlier.

Safety analyses were performed in all enrolled participants ($n = 400$), whereas the immunogenicity assessment population (per-protocol population) included 380 (95%) participants (186 in the BE-PCV14 group and 194 in the PCV13 group) who completed all study visits with data available for immunogenicity. Of the 20 participants who did not complete the study procedures, 14 migrated from the study area, 3 withdrew consent, 2 were lost to follow-up, and 1 received another vaccine as a booster dose (study schema is shown in [Fig. 1](#)). During the course of this study, no major protocol deviations were reported at any of the study sites. A few participants attended visits outside the window period, but these deviations were not significant and were duly reported to the ethics committees of the respective study sites. Data from these participants were included in both the safety and immunogenicity analyses.

Baseline demographic and anthropometric characteristics of the intention-to-treat cohort are summarized in [Table 1](#). Both the BE-PCV14 and PCV13 groups had a mean age at enrolment of 48 days, with comparable mean length (≈ 53 cm) and mean weight (≈ 4.2 kg). Female participants represented 48.0% of the BE-PCV14 arm and 43.5% of the PCV13 arm. Infants in both groups received DTwP-HepB-Hib, rotavirus vaccine, and IPV as part of the routine immunization schedule at Visits 1, 2, and 3.

Immunogenicity findings

The primary endpoint was the proportion of infants who achieved seroresponse at three pre-specified timepoints: Day 84 (post-primary series), Month 9 (pre-booster) and Month 10 (one month post-booster) in each vaccine group.

At the post-primary time point, seroresponse rates in the BE-PCV14 arm were comparable for 11 of the 12 shared serotypes, with serotype 18 C showing a numerically higher rate in the PCV13 group. At the pre-booster timepoint, BE-PCV14 seroresponse rates were higher than those of PCV13 for serotypes 3 (57.0% vs 44.8%), 6 B (90.9% vs 68.6%), 9 V (69.9% vs 59.8%), 18 C (55.9% vs 44.8%) and 23 F (70.4% vs 42.8%). One month post-booster (Month 10), seroresponse exceeded 88% in BE-PCV14 arm and 85% in PCV13 arm for every shared serotype, with virtually identical rates across all common serotypes. Detailed seroresponse data for each time point are illustrated in [Fig. 2](#) (bar graphs) and tabulated in [Supplementary Data Appendix 6](#).

Serotype-specific IgG geometric mean concentrations (GMCs) are shown in [Fig. 3](#) and tabulated in [Supplementary Data, Appendix 7](#). At Day 84 (post-primary series), BE-PCV14 induced GMCs ranging from 0.59 $\mu\text{g/mL}$ (95% CI: 0.52, 0.66) for serotype 3–8.46 $\mu\text{g/mL}$ (95% CI: 7.02, 10.18) for serotype 14; in the PCV13 arm, post-primary GMCs ranged from 0.54 $\mu\text{g/mL}$ (95% CI: 0.48, 0.60) for serotype 3–6.88 $\mu\text{g/mL}$ for

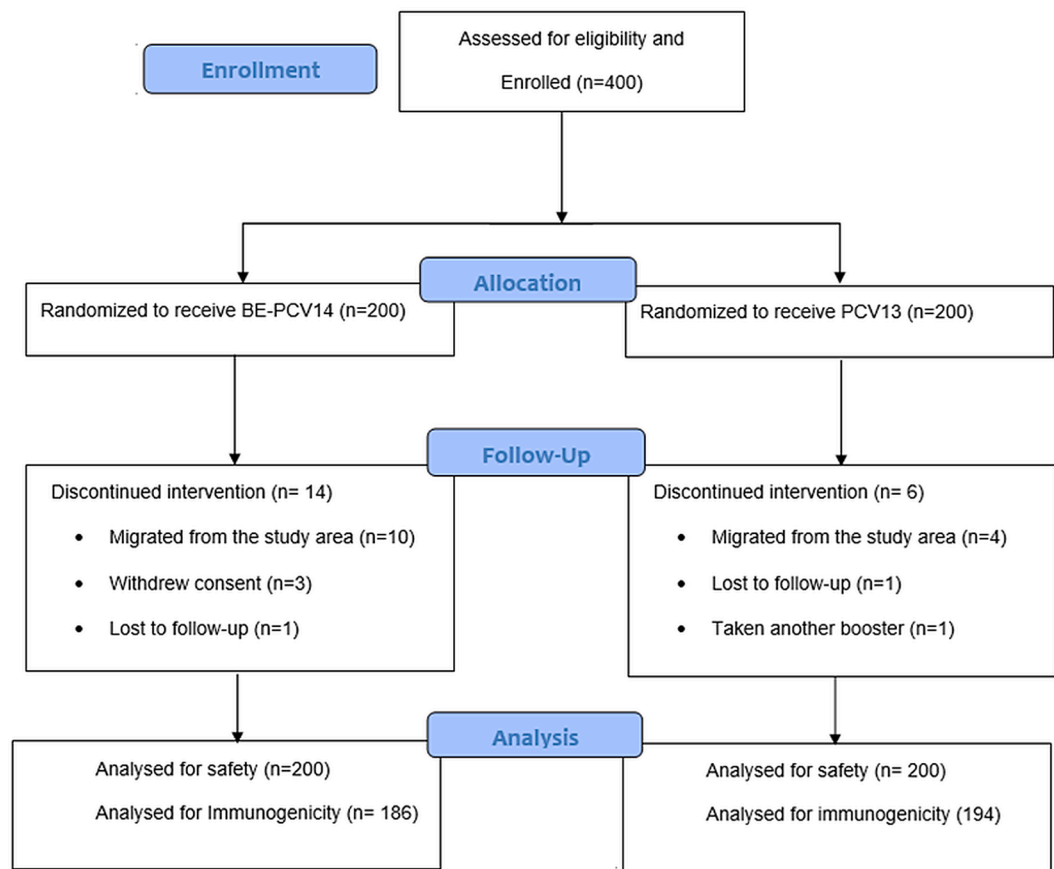


Fig. 1: Study Schema. (n: number, PCV: Pneumococcal conjugate vaccine, BE-PCV14: Biological E's 14 valent PCV, PCV13: Pfizer's 13 valent PCV).

serotype 19 F. By Month 9 (pre-booster), BE-PCV14 GMCs varied from 0.42 $\mu\text{g/mL}$ (95% CI: 0.37, 0.47) for serotype 3–4.89 $\mu\text{g/mL}$ (95% CI: 4.09, 5.83) for serotype 14; corresponding PCV13 GMCs ranged from 0.34 $\mu\text{g/mL}$ (95% CI: 0.30, 0.38) to 4.31 $\mu\text{g/mL}$ (95% CI: 3.62, 5.15). One month post-booster (Month 10), BE-PCV14 elicited GMCs of 0.88 $\mu\text{g/mL}$ (95% CI: 0.78, 0.99) for serotype 3 up to 10.06 $\mu\text{g/mL}$ (95% CI: 8.72, 11.62) for serotype 14; PCV13 GMCs ranged from 0.73 $\mu\text{g/mL}$ (95% CI: 0.65, 0.81) to 10.60 $\mu\text{g/mL}$ (95% CI: 9.30, 12.09) for serotype 14.

Reverse cumulative distribution curves for all shared serotype IgG responses at the post-primary and post-booster time points are provided in the [Supplementary Data \(Appendix 8\)](#).

Cross-reactive immunogenicity against serotype 6 A, not included in the BE-PCV14 vaccine, was evaluated to assess potential protection mediated by serotype 6 B. Following the two dose primary series, seroresponse rates for 6 A were 70.4% (95% CI: 63.5, 76.5) at Day 84 (post primary) and 69.3% (95% CI: 62.4, 75.5) at Month 9 (pre booster), rising to 93.0% (95% CI: 88.4, 95.9) at Month 10 (post booster). Corresponding IgG geometric

mean concentrations were 0.84 $\mu\text{g/mL}$ (95% CI: 0.68, 1.04) post primary and 0.96 $\mu\text{g/mL}$ (95% CI: 0.78, 1.17) pre-booster, increasing to 2.19 $\mu\text{g/mL}$ (95% CI: 1.82, 2.65) after the booster dose. Comparative immunogenicity data for serotype 6 A between BE PCV14 and PCV13 after the three-dose series are summarized in [Table 2](#).

Post hoc non-inferiority analyses showed that BE-PCV14 was non-inferior to PCV13 by both IgG GMC ratios ([Fig. 4a](#)) and seroresponse rates ([Fig. 4b](#)) for all 12 common serotypes, and for the two serotypes (22 F and 33 F, when compared with the lowest performing serotype of PCV13). These data are represented as forest plots in [Fig. 4](#).

Safety findings

Safety evaluations were predefined as secondary objectives. No adverse events (AEs) were observed within the first 60 min following any of the three vaccinations. Over the entire study period, 225 AEs were reported among 70 participants (35.0% [95% CI: 28.73, 41.84]) in the BE-PCV14 cohort vs 236 AEs in 77 participants (38.5% [95% CI: 32.03, 45.40]) in the PCV13 cohort. The

Characteristic	Statistics	BE-PCV14 (N = 200)	PCV13 (N = 200)	Overall (N = 400)
Age (Days)	n	200	200	400
	Mean	48.4	48.0	48.2
	SD	3.94	3.77	3.85
	Median (Q1, Q3)	49.0 (45, 51)	47.0 (45, 50)	48.0 (45, 51)
	Min	42	42	42
	Max	56	56	56
Length (cm)	n	200	200	400
	Mean	53.3	53.2	53.3
	SD	2.15	2.17	2.16
	Median (Q1, Q3)	53.0 (52, 54)	53.0 (52, 54)	53.0 (52, 54)
	Min	50	50	50
	Max	70	65	70
Weight (gms)	n	200	200	400
	Mean	4245.0	4233.1	4239.0
	SD	506.28	527.69	516.48
	Median (Q1, Q3)	4175.0 (3900, 4600)	4150.0 (3900, 4630)	4150.0 (3900, 4600)
	Min	3300	3335	3300
	Max	5600	5900	5900
Gender	Male	104 (52.00%)	113 (56.50%)	217 (54.25%)
	Female	96 (48.00%)	87 (43.50%)	183 (45.75%)
Nationality	Indian	200 (100%)	200 (100%)	400 (100%)

Percentages were calculated using respective column header count as denominator. n, subject count; N, sample size. ITT, intention-to-treat.

Table 1: Demographic and baseline characteristics—ITT population (N = 400).

vast majority of events were solicited and occurred within seven days of vaccination (Fig. 5a; Appendix 4).

The adverse events in the BE-PCV14 arm were most commonly injection-site pain (35 participants; 17.5%; 40 events), pyrexia (28; 14.0%; 35 events), injection-site swelling (12; 6.0%; 12 events), and erythema (11; 5.5%; 12 events). Among PCV13 recipients, pyrexia (37; 18.5%; 47 events), injection-site pain (34; 17.0%; 42 events), swelling (21; 10.5%; 24 events), and erythema (9; 4.5%; 9 events) were most frequently reported (Fig. 5b). A tabular overview of the overall AEs in this study are provided in [Supplementary Data, Appendix 5](#).

Two serious adverse events (SAEs) occurred in the BE-PCV14 group: one case of lower respiratory tract infection in an infant with a patent ductus arteriosus, and one case of urinary tract infection. Both events were adjudicated by the principal investigators as unrelated to the study vaccine and resolved fully. No deaths, or clinically significant abnormalities in vital signs or physical examinations were reported in either group. Overall, the safety profile of BE-PCV14 was comparable to that of PCV13.

Discussion

This multicentre, randomized, single-blind, phase III trial was the first to compare BE-PCV14 with PCV13 administered in a 2p + 1 schedule (two primary doses at

6 and 14 weeks followed by a booster at 9 months) in Indian infants.

Overall, infants who received BE-PCV14 achieved robust serotype-specific IgG responses comparable to PCV13, with strong priming post-primary (Day 84: >90% seroresponse for 8/12 shared serotypes and >80% for three). At the pre-booster timepoint (month 9), both vaccines showed expected waning, but BE-PCV14 demonstrated better persistence for several serotypes, with higher pre-booster seroresponse rates for 3, 6 B, 9 V, 18 C, and 23 F. Programmatically, stronger persistence through month 9 can be particularly meaningful in real-world settings where booster timelines can vary thereby helping buffer during the interval when disease risk remains clinically important.

Following the booster dose, the seroresponse rates exceeded 90% for most serotypes and met the *post hoc* criteria of non-inferiority for all 14 serotypes. The consistently strong booster *catch-up* across serotypes also supports that the 2p+1 schedule is effectively generating immune memory with BE-PCV14, reinforcing confidence in a schedule that India already delivers at scale.

Post-booster GMCs exceeded pre-booster levels for every serotype and surpassed post-primary GMCs for the majority of serotypes. A *post hoc* analysis demonstrated non-inferiority of BE-PCV14 to PCV13 for all 14 serotypes. Taken together, these findings imply that BE-PCV14 can be substituted into a 2p+1 program without

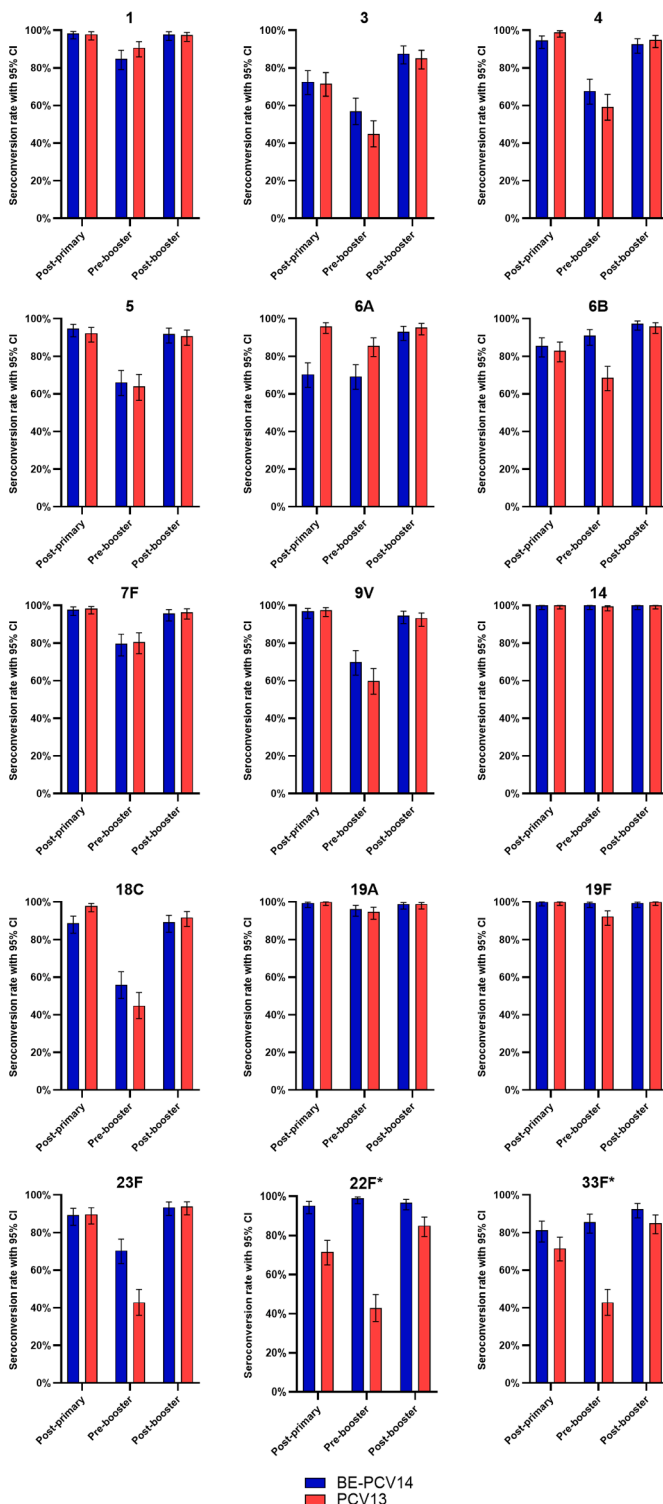


Fig. 2: Proportion of participants with serotype specific seroresponse rate—PP population (n = 380). (Data provided for 15 serotypes 1, 3, 4, 5, 6 A, 6 B, 7 F, 9 V, 14, 18 C, 19 A, 19 F, 23 F, 22 F & 33 F. BE-PCV14: 14-valent Pneumococcal Polysaccharide Conjugate Vaccine of

sacrificing the booster-driven protection that underpins durability and indirect effects in many mature PCV programs.

BE-PCV14 induced high seroresponse rates and IgG GMCs after the two-dose primary series and elicited robust booster responses against its unique serotypes 22 F and 33 F. Compared with the lowest-performing PCV13 serotype at the post-booster time point, BE-PCV14 achieved superior seroresponse rates and GMC ratios. These serotypes are now among the leading non-vaccine serotypes, accounting for 9–16% of invasive pneumococcal disease cases in children and adults in regions with established infant PCV programs^{22–24} and demonstrate an increased propensity to cause invasive disease following nasopharyngeal colonization in children.²⁵ Including 22 F and 33 F in a conjugate vaccine therefore represents a strategic approach to further reducing the global burden of pneumococcal disease. A practical implication is improved future-proofing against serotype replacement, adding 22 F/33 F may help maintain population impact as serotype ecology shifts after widespread PCV use.

In the current study, we also evaluated the cross-protective immunogenicity of BE-PCV14 against serotype 6 A, attributable to immune responses against serotype 6 B, which is included in the vaccine, owing to the high structural similarity between these two serotypes. The seroresponse rate of 70% following 2 dose primary series is consistent with the published 3 dose primary series data for the same vaccine.^{16,17} Seroresponse rates and IgG GMCs were sustained through the pre-booster time point without evidence of waning. Following the booster dose, seroresponse increased to 93.0%, comparable to 95.4% with PCV13, and IgG GMC rose from 0.84 µg/mL post-primary to 2.19 µg/mL, indicating a robust booster response. These results mirror those of Väkeväinen et al., who reported sustained immunogenicity through the pre-booster period and a robust booster response against cross-reactive serotype 6 A following PCV7-CRM vaccination in infants.²⁶

From a safety perspective, both vaccines were well tolerated with no safety signals identified. Adverse events were predominantly mild to moderate, solicited in nature, and occurred within the 7-day post-vaccination period. No adverse events were reported within 60 min of vaccine administration. The overall incidence of adverse events was comparable between groups: 35.0% in the BE-PCV14 group and 38.5% in the PCV13 group. The most frequently reported adverse

Biological E. Limited, PCV13: Prevenar13® (13-valent Pneumococcal Conjugate Vaccine from Pfizer), *For 22 F and 33 F, the rates in BE-PCV14 were compared with the lowest performing serotype rates in PCV13, Error bars shows 95% Confidential Interval).

events in both groups were pyrexia and injection-site pain, consistent with established safety profiles of pneumococcal conjugate vaccines. No deaths were reported during the study; two serious adverse events (SAEs) occurred in the BE-PCV14 group, both assessed by investigators as unrelated to the study vaccine. Clinical examinations, vital signs, and physical assessments throughout the study revealed no vaccine-related concerns. These safety findings align with safety results obtained from previous trials of BE-PCV14.^{16,17}

Multiple randomized trials and meta-analyses have assessed pneumococcal conjugate vaccines in 2p + 1 schedules.^{14,27–33} Overall, 2p + 1 elicits serotype-specific IgG and opsonophagocytic activity comparable to 3p + 0/3p + 1, with similar pre-/post-booster GMCs.²⁷ Effectiveness remains high: FinIP reported 92% protection,²⁸ England observed ~90% reductions with herd effects,^{29–31} and Netherlands/Finland showed ≥90% VE.^{32,33} Australia found fewer breakthroughs vs 3p+0.¹⁴ Immunologically, positioning the third dose in the 9–15 months of life appears advantageous: the systematic review cited earlier showed higher booster-induced GMCs when the final dose was deferred beyond six months, supporting longer antibody persistence.²⁷ Nasopharyngeal carriage studies also report equal or greater reductions in nasopharyngeal vaccine-type carriage after 2p + 1 compared with 3p + 0, reinforcing the potential for indirect protection.³⁴

Notably, not all evidence favours reduced priming in every epidemiologic context. WHO's 2025 position paper³⁵ notes that cross-country schedule comparisons (2p + 1 vs 3p + 0 vs 3p + 1) are subject to confounding because 2p + 1 evidence is dominated by high-income settings while many 3p + 0 data come from Africa, and some post-PCV carriage trends can be atypical in specific settings. In addition, for reduced-dose infant strategies (e.g., 1p + 1), meta-analytic findings summarized by WHO show that while vaccine-type carriage may be similar to 3-dose schedules, early post-primary IgG (and in some analyses certain serotype GMCs after the final dose) can be higher with 3-dose schedules, highlighting that fewer priming doses can trade off early direct protection for programmatic simplicity in some contexts. Additionally, in a recent Phase IV study, we have evaluated the safety and immunogenicity of BE-PCV14 in a 3p + 1 schedule demonstration robust boosting with a 4th dose administered in toddlers.³⁶

Taken together, global evidence complements our data and demonstrates that the 2p + 1 strategy provides robust, durable immunity and clinical effectiveness that are non-inferior to traditional 3-dose infant schedules. The 2p + 1 schedule also affords programmatic flexibility and potential cost savings for national immunisation programmes.

1 + 1 dosing schedules (UK example) and relevance to higher-valent PCVs: The United Kingdom changed

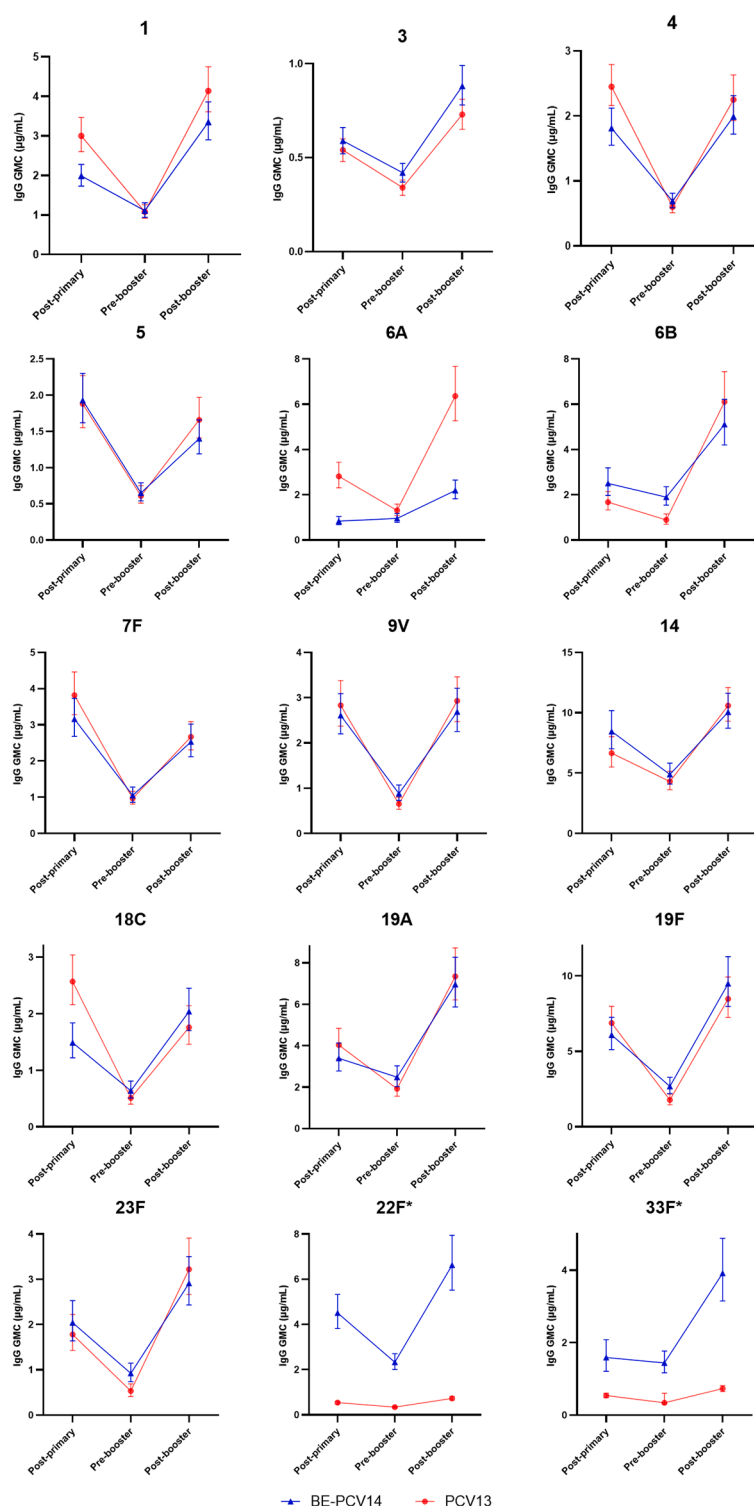


Fig. 3: Summary of Geometric Mean Concentrations (GMCs) of serotype specific anti-PnCPS IgG antibodies—PP population (n = 380). (*For serotype 22 F and 33 F, the GMC in BE-PCV14 were compared with lowest performing serotype GMCs in PCV13 at post-primary, pre-booster and post-booster time points).

Parameter	BE-PCV14 (n = 186)	PCV-13 (n = 194)
SRR (%)	93.0 (88.4, 95.9)	95.4 (91.4, 97.5)
IgG GMC (µg/mL)	2.19 (1.82, 2.65)	6.36 (5.27, 7.68)

Table 2: Serotype 6 A seroresponse rate (SRR) and Geometric Mean Concentration (GMC)—PP population (N = 380).

its infant PCV schedule on 1 January 2020 from 2p + 1 (8 and 16 weeks + 12-month booster) to 1p + 1 (12 weeks + 12-month booster). WHO's 2025 position paper³⁵ summarizes early post-implementation surveillance indicating stable IPD incidence across age groups (including infants and toddlers) in initial post-switch years, and cites national observational analyses of England's 1 + 1 program. At the same time, WHO's SAGE-commissioned 2025 systematic review explicitly frames 1p + 1 evaluation around disease incidence, immunogenicity, and nasopharyngeal carriage underscoring that reduced-dose policies should be evidenced and context-specific (e.g., coverage, timeliness, baseline transmission, and existing indirect effects). For higher-valent PCVs (including PCV14), a logical next step would be to generate bridging evidence initially immunogenicity/OPA and carriage before considering whether a 1p + 1 strategy could preserve both direct protection in early infancy and indirect protection at population level. Given India's higher transmission settings in many regions, assessing any 1p + 1 strategy would likely require particularly strong

assurance on early infant protection, booster timeliness, and robust surveillance infrastructure. An ongoing investigator-initiated trial is comparing the 1p + 1 and 2p + 1 schedule of BE-PCV14 with respect to immunogenicity and impact on nasopharyngeal carriage (CTRI/2024/12/077626). Notably, the recent WHO prequalification of BE-PCV14 (PNEUBEVAX 14®) enables potential procurement via UN agencies, which can be a key lever for affordability and supply security in lower- and middle-income settings.

While the results are promising, further post-marketing surveillance will be essential to monitor the long-term safety and real-world effectiveness of BE-PCV14 across diverse populations. Continued evaluation in different geographic regions and in high-risk subpopulations will be critical to fully characterize the vaccine's performance in routine clinical use. In particular, linking immunogenicity to real-world endpoints (IPD, radiologic pneumonia, and carriage/serotype replacement dynamics) will be important to demonstrate incremental public-health value from adding 22 F and 33 F in settings with evolving serotype ecology.

This study was limited in its modest sample size and descriptive design comparing BE-PCV14 and PCV13 in a 2p+1 schedule. Although the study was not statistically powered to detect between-group differences, *post hoc* analyses demonstrated that BE-PCV14 achieved non-inferiority to PCV13 for all 14 serotypes with respect to seroresponse rates (with *post hoc* power ranging from ~89% to 95%) and IgG geometric mean

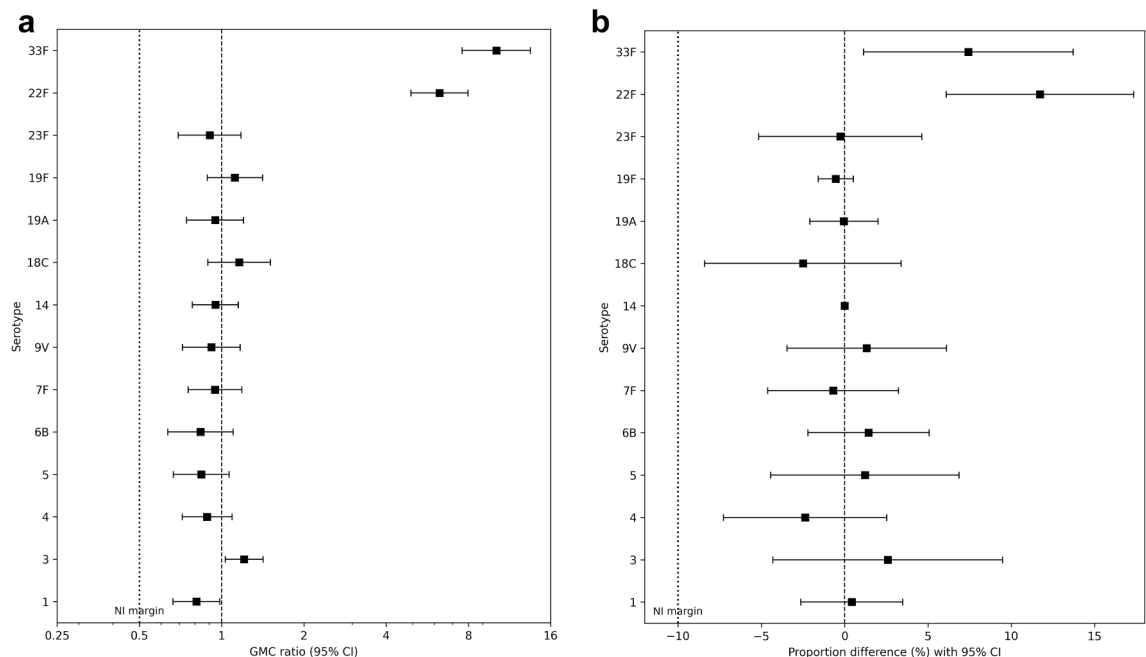


Fig. 4: (a) Forest plot for non-inferiority between BE-PCV14 and PCV14 IgG GMC Ratio at Day 300 (Post-booster). (b) Difference in seroresponse rate at Day 300 (Post-booster).

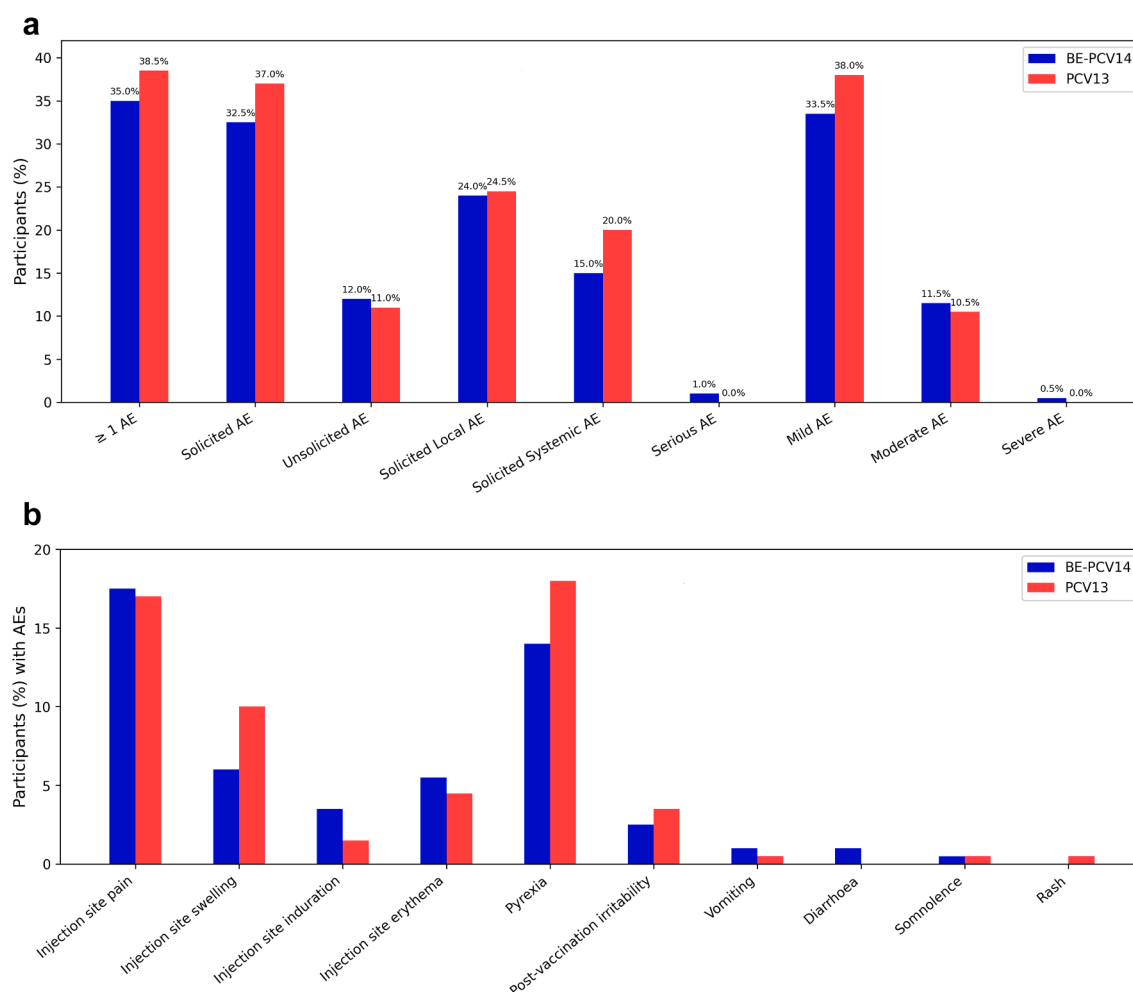


Fig. 5: (a) Overview of participants with adverse events after vaccination—Safety population (N = 400). (b) Solicited adverse events after vaccination—Safety population (N = 400).

concentrations following the booster dose. Another limitation was that all immunogenicity analyses were based on pneumococcal capsular polysaccharide (PnCPS) IgG concentrations and functional antibody activity via opsonophagocytic assay (OPA) was not assessed in this study. Nonetheless, data from the pivotal Phase III trial of BE-PCV14 using a 3p + 0 schedule demonstrated comparable OPA titres with PCV13. A further limitation is the use of the ≥ 0.35 $\mu\text{g/mL}$ anti-capsular IgG threshold as a serologic correlate, as this cut-off is an aggregate, serotype-nonspecific population benchmark and may not uniformly predict protection across serotypes, outcomes (IPD vs pneumonia), or schedules, potentially affecting interpretation of seroresponse comparability. This study is limited in its applicability as it was only conducted in India. Studies of BE-PCV14 in other geographies are presently being planned.

In conclusion, BE-PCV14 demonstrates strong immunogenicity, both post-primary with good persistence and post booster vaccination in 2p + 1 schedule and an acceptable safety profile in Indian infants. Its inclusion of additional serotypes not present in 10- and 13- valent vaccines offers the potential for enhanced protection against IPD, supporting its consideration as a valuable tool in the global fight against pneumococcal disease. BE-PCV14 also induced robust cross-protective immune response with strong booster response against serotype 6 A. Future real-world effectiveness studies will be essential to elucidate the performance of BE-PCV14 when administered in a 2p + 1 immunization schedule. With the recent WHO prequalification of BE-PCV14, the use of this vaccine is likely to expand beyond India through global procurement mechanisms, generating real-world evidence on immunogenicity, effectiveness,

and impact on pneumococcal carriage under routine programme conditions.

Contributors

Subhash Thuluva: Conceptualization, Methodology, Formal analysis, Data curation, Project administration, Resources, Supervision, Writing—original draft, review & editing, Decision to submit the manuscript. Ramesh V. Matur: Conceptualization, Funding acquisition, Formal analysis, Data curation, Writing—review & editing. Subbareddy Gunneri: Methodology, Project administration, Resources, Writing—review & editing. Siddalingaiah Ningaiah: Formal analysis, Visualization, Writing—review & editing. Vijay Yerroju: Data curation, Writing—review & editing. Rammohan Reddy Mogulla: Data curation, Formal analysis, Writing—review & editing, Data Accessing and Verification. Chirag Dhar: Formal analysis, Visualization, Writing—review & editing. Kamal Thammireddy: Methodology, Validation, Writing—review & editing, Data Accessing and Verification. Piyush Paliwal: Formal Analysis, Writing—review & editing. Anand Kawade, Pradeep N, Jog Pramod, Manish Narang, B.S. Chakravarthy, Prashanth M.V.: Investigation, Writing—review & editing, Data Accessing and Verification.

Data sharing statement

All relevant data are included in the manuscript. Additional reasonable requests may be directed to the corresponding author.

Declaration of interests

ST, RVM, SG, SN, VY, RRM, CD, KT, and PP are or were employees of Biological E. Limited. None of the authors hold, or have held, any equity or stock in the company. AK, PN, JP, MN, BSC, and PMV were principal investigators on the study and received research funding support from Biological E. Limited. All authors declare no other relevant conflicts of interest.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.lansea.2026.100746>.

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